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NOTE this is a successor agreement to "Applications of Electric Fields in Microgravity Combustion Science (NNX07AB55A)," per D. Stocker, NASA Glenn Research Center.
This report summarizes recent findings from a joint project between the University of California, Irvine and Imperial College, London on the use of electric fields in combustion systems, including preparations for experiments in microgravity on the International Space Station (ISS) that will elucidate the fundamentals of chemi-ionization and the potential applications of ion-driven winds for combustion sensing and control.

Activities in this review period include:

1.) Determining the dynamic response of compact high-voltage power supplies for electric field generation (both for the ISS experiments and for potential terrestrial applications), and the flame's behavior when subjected to these power

Task Description:	supplies, 2.) Determining the effects of electric fields on flames that are impinging on quenching surfaces, including controlling the release of carbon monoxide (CO) and directing thermal convective flux, 3.) A detailed diagnostics study on CH* chemiluminescence in sooting flames both for use in analyzing the ISS xperiments and for quantifying a possible relationship between CH* emission and ion current, and 4.) Developing a ground-based apparatus to provide a refined understanding of how changes in hardware performance may affect the optimal test matrix for flight experiments. Our understanding developed in these areas may improve kinetic mechanisms for diffusion flame modeling (including charged species), help clearly identify the risk of CO emission from quenched flames, and help determine if electric fields can actively control the release of carbon monoxide from flames. Task Description: The overall goal of this project is to understand the behavior of chemi-ionization in flames and to demonstrate the use of this understanding to manipulate combustion in beneficial ways (including enhancing flame stability and reducing flame soot). The project uses detailed measurements of electrical properties of flames in several ways. First, we can create a neutral wind (ion driven convection) locally at the flame front; this is one of the rare and important action-at-a-distance effects on flames that can be used as a localized actuator. The effect of the wind has been demonstrated with a candle flame in parabolic flight based microgravity and in a spherical cage drop tower, but there has been no systematic study of the ion wind effect in the absence of the confounding effects of the approximately equal influence of buoyancy. Second, we can use electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions. It is possible, therefore, to modify soot formation and to use charges attached to soot to affect soot transport. By adjusting the soot trajectory through the reaction zone it is possible to affect the oxidation of soot to prevent its eventual release or to modify the residence time of soot in different zones of the flame. It is also possible to use the neutral ion wind to change the mixing behavior of the flame and reduce soot in this way. Finally, we can use the electrical properties to characterize the flame – there is a quantifiable relationship between the carbon content of a fuel and the ion current it can produce at saturation. This relationship is already exploited in a typical flame ionization detector (FID) analyzer for hydrocarbon emission where the carbon is burned in a hydrogen flame to create an ion current in proportion to the carbon influx. In direct flame interrogation, however, there are several confounding variables that can affect the ion current, including flame temperature (which generally affects the O atom concentration) and the reaction zone volume. Flame temperature can in turn be affected by levels of premixing, diffusion rates, fuel composition, and radiative loss. It is possible, therefore, to determine flame character from the ionization curve response. Task Significance: Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The control of combustion has the potential to improve efficiency and reduce emissions from burning fuels. Since high power density often requires combustion, these improvements will be important no matter what the fuel source. Electric fields acting on flames have the potential to aid in combustion control. In addition, electrical properties of flames can identify poor performing boiler flames that release poisonous carbon monoxide. Our studies show that a flame's electric signature can capture incipient quenching before dangerous emissions result. Understanding the links between electrical character and flame behavior may allow improved sensing of poor performing combustion systems.
	Flame Dynamic Response to High-Voltage Power Supplies--A portion of our research this year was directed toward characterizing a pair of High Voltage Power Supplies (HVPS) manufactured by UltraVolt because these supplies are the proposed high voltage source for a future space flight experiment aboard the ISS. Since our laboratory experiments were completed using a larger more robust power supply from TREK, it was valuable to evaluate possible differences. The results show thatTREK takes a few milliseconds (order of magnitude) rise time, while the HVPS takes approximately 100 ms. This could be an important issue since we are expecting the flame to have a response time on the order of 100 ms as well, and so determining electric field dynamics can be complicated by the slow rise time of the HVPS. By characterizing the power supply dynamics we can potentially deconvolve the flame response analytically. It is also possible that the power supply dynamics will not be as dramatic for ramps and smaller step changes, which are also important planned ISS experiments. On the ramp response, a 1kV square wave signal with different cycle times, 200 ms, 100 ms, and 50 ms is used. The results show that the HVPS exhibits about 0.2 kV overshoot after it reaches the desired 1 kV and then ripples down toward 1kV. On the TREK side, the overshoot is much smaller and can be neglected (about 10V). Although there is an overshoot, in the mean value the HVPS maintains agreement with the requested voltage for 200 ms and 100 ms cycles, and the response starts delaying for 50 ms steps. The TREK is able to work with accurate response for nearly all conditions and serves an ideal baseline result for comparison. Another response when asking for 5kV and back to 0V (in a large triangle profile) with 500V increments taken stepping up and down is also tested. This series mimics what will be a voltage-current characteristic curve sweep in the E-FIELD experiment. Results present that with the requested step getting smaller, the HVPS responds slower with higher cycle rates. The voltage overshoots and starts to fall before the value has converged to the request value; the step shape is no longer a square in both 10 ms, and 7.5 ms cases. The TREK can reach the requested voltage, and its value converges immediately. Its voltage is a little higher than the requested voltage but the difference is not significant. As mentioned before, the HVPS shows good agreement on 100 ms timescales. However, this timescale is approximately

same as that associated with the buoyant flows, making it difficult in 1g to distinguish the effects of the power supply and that of the flame. Fortunately, with a continuous measure of the current and voltage we expect to be able to deconvolve the effects. Some zero-g drop tower tests are planned to confirm this deconvolution approach. The ion current is one of the significant characteristics describing the flame in an electric field since it reflects the resistance to current flow at a given voltage. The maximum or saturation ion current for the small diffusion flame of these experiments is on the order of a few micro-amps. The steady ion current with five different flow conditions, starting at 10 cm/s with 5cm/s as increments, are measured and plotted as VCC (Voltage-Current Characteristic) curves. The VCC curves shows the typical distinct three regions of the ion current behavior in a flame. At first, the ion current rises (or falls) following a quadratic curve with increasing field strength. Then it reaches a plateau region and stops rising. With increasing field strength, the ion current rises again, following a parabolic curve until the flame blows off (upward directed ion driven wind – negative mesh) or is extinguished (downward directed ion driven wind – positive mesh). Those three regions are called the subsaturation region, the saturation region, and the supersaturation (or secondary ionization) region, respectively. The supersaturation region is often complicated by corona discharge effects from the burner and mesh so it often better to avoid the extreme voltages when trying to understand the electrical aspects of the flame alone.

The results demonstrate that the power supplies correspond to each other in the saturation region. However, there are considerable differences in the other regions. The ion current measured using the HVPS rises (or falls) with smaller field strength in the subsaturation and supersaturation regions. Therefore transition points in between the regions appear at lower field strength with the HVPS. It appears that the presence of the current sink component in the power supply affects the ion current. That trend is more significant under the higher flow condition. The power supplies are in good agreement with each other under the lower flow speed condition. This is a promising finding for the zero-gravity studies planned because electrical effects and their competition with buoyancy are much more pronounced when there is lower forced convective momentum. The flame response in terms of ion current is examined following a step input from 0 V to 5 kV. The initial spike observed in the TREK case corresponds to the capacitance of the burner/mesh electrode system. The current then decays to a steady value representative of the flame/electrode gap space. This phenomenon was characterized fairly extensively in prior work in this project. The rising ion current after the initial spike is likely due to the flame finding a new accommodation with the ion driven wind. The flame then settles to a steady condition after 30-40 ms. The HVPS results show a much less prominent spike because the slower response time does not engage the small system capacitance. It is very interesting to note, however, that the settling time of the ion current, and its ultimate value matches very well the results obtained when using the TREK. The insensitivity results because the requested voltage of this measurement is in the saturation region where the ion current does not rise with electric field strength. Therefore, the electric response is largely unaffected by the power supply response time. This a fortuitous condition that provides an important opportunity to maximize the understanding of flame behavior even when using a slow responding power supply. That is, operating with step changes between saturated regions can create more rapid response changes in the ion current than is strictly available in the voltage.

While there is experimental verification that the ion current response between the two power supplies is quite similar in time, the connection between this ion current and the flame behavior is still needed. Visualized flame response following steps of voltage using the TREK and the HVPS are captured with a camera filming at a rate of 300 fps. The mesh polarity is initially negative then a positive step voltage is applied, producing an increasing voltage range from -9 kV to +9 kV.

The results show that the TREK takes about 20 ms while the HVPS takes 130 ms to reach +9 kV. Generally, the flame is stabilized while the high voltage is applied. When the mesh is negative, the flame is extended vertically upward, and it becomes thinner. Conversely, while the mesh is positive, a downward wind is imposed, opposing natural convection. While only qualitative, it appears that with a simple time stretching the flame character matches between the two power supplies. This is slightly different from the ion current finding, which would have indicated that the flame response time should be unaffected. The difference is that in the -9 kV to 9 kV test the step is not into a saturated ion current condition but between two conditions in the supersaturated regions. In this circumstance, there will be sensitivity in the ion current. In any case, the still images are slightly misleading, and the high speed movies show more dynamic processes. In the TREK case, for example, the video shows that the flame is stabilized just 10 ms after the polarity is reversed. The flame transitions smoothly with little fluctuation from the narrow stretched shape with the upward wind to the flattened shape with the downward wind. On the other hand, when using the HVPS, the flame takes more than 100 ms to stabilize, and it continues to exhibit oscillations. With the slower responding power supply, the flame development seems separated into two stages where the flame reacts quickly at first and then slowly widens. This behavior corresponds to the electric response of the HVPS in step input responses. The oscillations are probably caused by the high voltage ripple in the output. The ripple does not affect the steady state ion current results because they are average values. Further study is needed but the dynamic differences noted above are consistent with considering the flame an active element of the system with a response time on the order of 30 ms. In the TREK case, the flame finds itself instantaneously facing an electric field condition that it then accommodates with a self-consistent change in shape and ion current generation. In the case of the HVPS the environment conditions (i.e., the electric field imposed) is continually changing as the flame tries to ccommodate the variation. This leads naturally to a more unstable behavior, particularly when the field is coupled with the ion current response. Again, further study in 1-g and zero-g will help sort out these effects.

UCI Coflow Burner--All ground-based experiments were performed using a coflowing burner that is dimensionally similar to the proposed ISS burner. The burner consists of two stainless steel plenums threaded together. Fuel, introduced in the bottom plenum, is passed through a stainless steel tube (2.13 mm ID) and exits the top surface fully developed. Air entering the top chamber passes through flow straightening beads and a honeycomb mesh. The air exits the top surface with a top hat flow profile at the exit. A variable high-tension power supply connected between the burner and downstream mesh electrode produces the applied electric field. The electric field distribution is deformed by the flame’s space charge. Regardless, defining an applied field strength, E = V/d, is a useful approximation when describing experimental results. Ion current is computed using Ohm’s law and the measured potential drop across a shunt resistor placed between the electrically isolated burner and the building ground. An acrylic chamber shields the flame from room currents.

Effects of Electric Fields on Impinging Flames--The ultimate practical aim of this series of experiments is to reduce the incidence of CO poisoning caused by the quenching of natural gas flames in malfunctioning burners. Concern about CO emission initially arose after we observed appreciable concentrations of CO within the luminous boundaries of our flames during experiments conducted at Imperial College, London, under the direction of Professor Felix Weinberg. These results prompted the question of how much carbon monoxide would escape into the atmosphere if quenching

disrupted the flame. We studied CO release for a variety of quenching surfaces, using an exhaust system that aspirated the emerging product gases, along with much ambient air. In addition to our efforts at Imperial College, parallel studies are underway at UC Irvine, studying the properties of partially quenched flames. This portion of our research is aimed at understanding the electrical behavior of a flame as a plate is incrementally lowered toward it by measuring the corresponding ion current. The study explores the concept of detecting flame impingement electrically without visual inspection. Previous work has shown that ion current sharply increases and finally saturates when lowering the plate toward the jet flame exit. The ion current decreases as the flame temperature and ion production fall. Further details on this topic appear in the publications (Chien and Dunn-Rankin, 2013; Chien et al., 2013).

Task Progress:

We continued the work to study the phenomena occurring on the quenching plate as the flame approached it. Different quenching surface materials have various emissivities and thermal properties that lead to different equilibrium temperatures. Preliminary research involves identifying the thermal emission from the plate at different heights when lowered towards the flame (or when the flame sheet is interrupted). In order to understand the thermal phenomenon, a FLIR camera is used. The FLIR camera uses a thermography technique for taking thermal images. The test ran with a stainless-steel plate that had been painted with a layer of non-gloss heat resistant paint to prevent reflections. The FLIR camera was located above the plate while a flame quenched underneath the plate. The microgravity relevance of this work is that convective heating in microgravity will depend on jet convection and ion-driven winds, rather than the combination of these flows and natural convection. To be clear, in this research all of the definitions of positive or negative electric field are based on the polarity of the burner relative to the downstream plate electrode. At 9 mm, the plate is very close to touching the outer dim luminous region of the zero-field flame but it does not reach the brightest flame front tip. For this plate location on the positive field side, the electric field drives positive ions and the ion wind toward the plate so that the flame is pulled up and the top of the flame narrows. The luminous zone at the flame front is brighter, presumably because the ion drift wind (also called Chattock wind) has entrained more oxidizer and shifted the reaction zone. When the flame is in positive field strength 6kV/0.9cm, the flame quenches. On the negative field side, the flame is pushed down gradually as the negative field strength grows. The downward ion wind spreads the flame and opens it. As in the case of the positive field, the flame quenches at approximately 6kV for the negative field but the extinguishment process is different. For the negative field, an arc develops between the plate and burner, and the arc disrupts the flame sufficiently to quench it. Similar phenomena can be seen when the plate is located at 5 mm, except that the flame is already open with zero field. The flame opens wider with increasing negative field strength, with the bottom rim of the flame close to the burner exit. This opening behavior results from downward directed positive ions producing a downward blowing wind when the burner is negatively charged.

To determine the effects of surface flame quenching on the electrical properties, the downstream plate is moved to different heights above the burner as the voltage is varied. At each different height of the plate above the burner, the range of electric field that is able to operate on the flame without extinguishing it or without generating an arc varies. The limits of the electric field become narrower as the gap between the plate and the burner decreases.

CO detection with high voltage control--Measurement of CO released from the flame with an electric field applied at different heights between the burner and the plate provides an indication of the potential for CO emission control of quenching flames.

The experiments are conducted for plate spacing from 3mm to 7mm, where the amount of CO generated is over one hundred ppm. The results show that CO concentration is affected significantly by the plate to burner quenching distance. Interestingly, CO concentration also varies with electric field. Normalized curves of this effect at each height with each measured value divided by the appropriate zero-field CO concentration is plotted. As the negative field grows, the CO level rises. In the positive field side, CO does not show a significant change but it still grows slightly just before falling near the breakdown or extinguishment limit.

The normalized results show a consistent trend that can possibly relate to unsteady fluctuations in the combustion driven by the electric field. With the plate progressively quenching, it is possible to estimate that the average CO concentration rises at larger negative field strength compared to the no-field case. There still exist mysteries on the effects of positive electric field strength to uncover any feasibility for CO control, but these preliminary findings suggest that electric field control near a quenching plate will have relatively minor positive influence on CO reduction. Electric fields may therefore be more important for sensing and for heat flux control of flames near surfaces.

Temperature measurement – FLIR camera--Temperature and species concentration measurements in flames that have an electric field applied have been a challenge for exploring local property distribution because probes of any kind disrupt the electric field significantly. The camera (FLIR SC620) measures radiative emission in the range from 7.5 to 13 microns, and it is situated directly above the quenching plate at normal incidence (to avoid complications associated with Lambert's cosine law). The camera is controlled by its native commercial software (ExaminIR). The quenching plate is painted with black paint and is considered a blackbody object. This assumption has been confirmed as reasonable using thermocouple temperature comparisons at a few spot locations.

The RGB image taken at thermal equilibrium shows a well-distributed temperature change on the plate. The temperature shown in the image does not represent a local temperature of the plate or the flame temperature. It is measuring the temperature of a thin layer of the heat resistant spray paint. The overall heat transfer includes forced convection of hot flame products onto the bottom surface of the plate, conduction through the plate, and then to the paint, the heat convection out of the surface into the air, and radiation from the surface, as the system reaches thermal equilibrium. Based on straightforward heat transfer analysis it is easy to conclude that the radiation from the paint provides a reliable measure of the impinging heat from the flame on the backside of the plate. It feasible, therefore, to use the FLIR camera to characterize a quenching flame and to acquire a temperature distribution using 2-D image technology. The thermal was recorded 60 seconds after changing the electric field, by which time the temperature is steady. The pressure drop during the experiment can affect the flow rate as a result of the small jet impinging on the plate, and the different flow rate affects slightly the temperature measured from the infrared camera, showing that about 1 ml/min change in the maximum plate temperature 2 Celsius degree. Therefore all experiments conducted with IR imaging experiment use a fuel flow rate of 37.7ml/min (which is 20m/s) ± 0.6ml.

Measurement at various heights (3, 5, 7, 9, 11 mm) show that temperature decreases with the field strength. Maximum temperature of each condition at each height and field strength is recorded, and the relative maximum temperature at zero electric field at each distance between the plate and the burner is also plotted. On the positive field strength side, the maximum temperature increases quickly in the beginning at 2kV/cm and keeps rising with the field strength applied. The reaction zone is narrowed due to the ion wind and the flame's hot gas being pulled up toward the plate. Both of

these effects produce an increased local heat flux. On the negative field strength side, the reaction zone of the flame is shifted wider with the increasing of the field strength. Therefore, the maximum temperature decreases. The curve shows that the reducing trend initially decreases with similar slope until -3kV/cm, and then drops more rapidly with more negative field applied. As the plate reaches 3mm, the lowest limit of its position, the electric field effect on the maximum temperature of the plate does not vary as dramatically as for plate positions that permit more electric field influence on the flame. This is because the plate is so close to the flame that the reaction zone of the flame has already opened widely. That is, the jet momentum modification caused by the proximity of the quenching plate is more substantial than the momentum modification caused by the ion driven wind. Since we have seen that the ion-driven wind effects decrease with decreasing plate height and the physical effects of the plate increase with decreasing plate height it is clear that there will be a trade-off between these two influences. These results need more evidence and a theory to explain how the electric field is able to control and manipulate the flame's heat flux to the plate.

CH* Chemiluminscence Measurements in Sooting Flames--The utility of chemiluminescent emission as a diagnostic tool is compromised in flames containing additional background radiation, particularly from soot. That is, a measurement of CH* chemiluminescence, even when using a narrowband interference filter, is obscured by incandescent particles in a sooting environment. The increased noise from soot affects the use of chemiluminescence as a quantitative marker and limits numerical validation of gas phase chemistry and soot oxidation mechanisms in these regions. This project effort utilized a filtering method employing three narrowband optical filters in order to subtract soot interference, and produce a soot-free CH* mage. Most of the key aspects of the method and findings were reported in the last annual report. The progress was refinement of the method, a conference paper and a journal paper that contain the details. Ultimately, by using the twin-filter method, the effect of soot temperature on the subtraction of soot luminosity was reduced below 1%. Developments toward Refined Test Matrices and Analysis for ISS Experiments--A ground-based experiment is being designed and built at UCI so that tests can be performed in preparation for analyzing ACME's E-Field Flames flight experiments. A single burner is being constructed to match the ISS system. The burner will act as both a co-flow and gas jet burner with interchangeable fuel tubes matching the 1.3 and 2.1 mm gas jet nozzle diameter requirement. The mass flow controllers (MFCs) units being used to supply the fuel, air, and N 2 gas streams to the burner are SFC 2010 from Semiflow Engineering, matching the performance of those selected for the ACME CIR insert. These three MFCs are being calibrated and programmed to supply the flow rates stated in the E-Field test matrices in the ACME SRD. While not ideal (as described earlier), the two Ultravolt 20A12 power supply units chosen for the flight experiment will be used to power the electrode mesh and burner electrode for the preliminary experiments so we can prepare a data analysis process that minimizes uncertainty in ion current dynamics. The electric circuit and data acquisition for the entire experiment is being designed to incorporate the control of the three MFCs and dual Ultravolt power supply units, as well as acquiring measurement data for electrode voltages and ion current.

Imperial College, London--The ultimate practical aim of the experiments begun at Imperial College was to reduce the incidence of CO poisoning caused by the quenching of natural gas flames in malfunctioning burners. The work examined the interaction of a flame with a quenching surface and the potential for using electric fields to detect the proximity of the surface and to control the CO release by inducing an ion-driven wind at the appropriate time. Sadly, Professor Felix Weinberg, the lead investigator at Imperial College and the pioneer in electrical effects applied to flames, passed away in December of 2012. The flame quenching studies (and they have been described above) have now been moved to UC Irvine.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
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Abstracts for Journals and Proceedings	Karnani S, Dunn-Rankin D. "Visualizing CH* Chemiluminescence in Sooting Flames." 8th US National Combustion Meeting, Park City, Utah, May 19-22, 2013. 8th US National Combustion Meeting, Park City, Utah, May 19-22, 2013. , May-2013
Abstracts for Journals and Proceedings	Chien YC, Dunn-Rankin D. "Electric Field Effects on Carbon Monoxide Release from Impinging Flames." 8th US National Combustion Meeting, Park City, Utah, May 19-22, 2013. 8th US National Combustion Meeting, Park City, Utah, May 19-22, 2013. , May-2013
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